

RCF Section 0 — Algebraic Foundation and the Two Primitives

Rewritten Canonical Form — v2.1 (Draft)

Preamble — What This Section Contains and Why

This is a structural rewrite of Section 0 against the reordered emergence chain. The v1.0 merge placed the fracture operator $\hat{F}$ (§0.3), the Reconciliation Propagator R_t (§0.4), the Convergence Theorem (§0.5), the thin/full split of A_{phy} (§0.6–0.7), the reduced algebra and Dirac bracket (§0.7b), and the Reconciliation Principle (§0.8) all within Section 0 — before causality (§1) and before locality was named as a primitive at all. That ordering created three structural problems: (i) the cross-extension network operator Π_{net} (Def 0.3.8) forward-referenced K_ω from §2; (ii) the reduced algebra A_{red} and its master constraint $\hat{M}_{\text{red}}$ were referenced by §4 but never defined, because the Tier 1/Tier 2 distinction requires the fracture machinery that didn't exist yet; (iii) the Reconciliation Principle was stated as a variational principle on burden before burden's physical interpretation (fracture cost) was available.

The rewrite strips Section 0 to its actual foundational content: the relational algebra, admissibility, the GNS construction, the algebraic definitions of $\hat{M}$ and $\hat{F}$, and the naming of causality and locality as structural co-primitives grounded in the Lie-Jordan decomposition of the algebra. Everything that depends on the fracture — R_t , the Convergence Theorem, A_{red} , $\hat{M}_{\text{red}}$, the Dirac bracket, burden as a physical quantity, and the Reconciliation Principle — moves to Section 1 (Open Expansion, Fracture, and the Reconciliation Propagator), where the fracture machinery makes these objects definable without circularity.

The test for this section: does it contain anything that depends on fracture, sub-Hilbert spaces, burden, R_t , A_{red} , $\hat{M}_{\text{red}}$, the Dirac bracket, or the Reconciliation Principle? If yes, the reordering has failed. If no, the architecture is acyclic at the foundational layer.

§0.0 Purpose of the Foundation

The Reconciliation Causal Framework (RCF) derives all physical structure — spacetime, fields, particles, gravity, probability, cosmology — from a single primitive: relations subject to constraints. Section 0 establishes the algebraic foundation on which every subsequent section rests. It does not assume a background manifold, a pre-existing Hilbert space, an external time parameter, or a measurement postulate. Instead it constructs the kinematic algebra, the admissibility condition, the GNS representation, and names the two structural co-primitives — causality and locality — whose complementarity drives the entire framework.

The central architectural commitment of this section is the two-primitive picture, grounded in the Lie-Jordan decomposition of the algebra. Every unital complex $(*)$ -algebra A carries two genuinely independent algebraic structures: the Lie structure $[A, B] = AB - BA$ (antisymmetric, capturing ordering) and the Jordan structure $A \circ B = \frac{1}{2}(AB + BA)$ (symmetric, capturing co-occurrence). These are mathematically independent — neither is

inferable from the other (Jordan-von Neumann-Wigner, 1934). Causality is the structural principle grounded in the Lie algebra A_L ; locality is the structural principle grounded in the Jordan algebra A_J . Neither reduces to the other. Their complementarity — the joint compatibility of Lie and Jordan, enforced by the associativity constraint $AB = \frac{1}{2}[A, B] + A \circ B$ — is the reconciliation principle, formally stated as a variational principle in Section 2.

A reader of the v1.0 merge will note that this Section 0 is significantly shorter than its predecessor. This is by design. The objects that were prematurely placed in v1.0 Section 0 — the Reconciliation Propagator, the Convergence Theorem, the thin/full split, the reduced algebra, the Dirac bracket, the Reconciliation Principle — all depend on the fracture of the Hilbert space into sub-Hilbert spaces, which requires the Open Expansion machinery of Section 1. Placing them here would create forward references and undefined objects. They are deferred to Section 1, where the fracture makes them definable.

§0.1 The Relational Algebra and Its Lie-Jordan Decomposition (Layer L — Linear)

****LAYER L · LINEAR****

Source: Front.pdf §0.1.2, Con.pdf §0.1, Section 0 2 §0.1. *Epistemic tag:* [Established].

The foundation begins with a single primitive: a unital complex $(*)$ -algebra A . Elements of A represent all possible relational operations; the involution $*$ represents the algebraic adjoint. No commutativity, no topology, no norm, no Hilbert space, and no spacetime is assumed at this stage. The algebra is purely algebraic: it carries addition, scalar multiplication, an associative product, a unit 1 , and an involution satisfying the standard $(*)$ -algebra axioms.

Definition 0.1.1 — Kinematic Algebra

> **DEFINITION 0.1.1 — Kinematic Algebra** [Established]

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> Let A be a unital complex $(*)$ -algebra with unit $1 \in A$. The involution $*$: $A \rightarrow A$ satisfies, for all $A, B \in A$ and $\lambda \in \mathbb{C}$:

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$$> (A^*)^* = A \quad (0.1.1a)$$
$$> (A + B)^* = A^* + B^* \quad (0.1.1b)$$
$$> (\lambda A)^* = \lambda^* A^* \quad (0.1.1c)$$
$$> (AB)^* = B^* A^* \quad (0.1.1d)$$
 $\succ$

> No commutativity, topology, norm, or Hilbert-space structure is assumed at this layer. The algebra is the sole primitive.

Definition 0.1.2 — The Lie-Jordan Decomposition

Every unital complex $(*)$ -algebra A carries two genuinely independent algebraic structures, obtained by decomposing the associative product AB into its antisymmetric and symmetric parts.

> **DEFINITION 0.1.2 — Lie-Jordan Decomposition** [Established]

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> Let A be a unital complex $(*)$ -algebra. The product AB of any two elements decomposes as:

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> $AB = \frac{1}{2}[A, B] + (A \circ B)$ (0.1.2)

>

> where:

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> - **The Lie bracket:** $[A, B] := AB - BA$ (0.1.2a)

> Antisymmetric ($[A, B] = -[B, A]$), capturing *ordering*. The Lie bracket equips A with the structure of a Lie algebra, denoted A_L .

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> - **The Jordan product:** $A \circ B := \frac{1}{2}(AB + BA)$ (0.1.2b)

> Symmetric ($A \circ B = B \circ A$), capturing *co-occurrence*. The Jordan product equips A with the structure of a (special) Jordan algebra, denoted A_J , satisfying the Jordan identity $(A \circ B) \circ (A \circ A) = A \circ (B \circ (A \circ A))$.

Remark 0.1.3 — Lie and Jordan Are Independent. The Lie and Jordan structures are mathematically independent. Neither is inferable from the other: there exist algebras with the same Lie structure and different Jordan structure, and vice versa (Jordan-von Neumann-Wigner, 1934). This independence is the algebraic foundation of the framework's two-primitive picture (§0.4). The Lie structure grounds causality (the "when"); the Jordan structure grounds locality (the "where"). Because they are algebraically independent, causality and locality are genuinely separable co-primitives.

Remark 0.1.4 — Associativity as the Compatibility Constraint. The decomposition (0.1.2) is not merely algebraic; it is the compatibility constraint between Lie and Jordan. To specify the full product AB — the full relation between A and B — one needs both the Lie part (the ordering) and the Jordan part (the co-occurrence), and these must fit together consistently. The associativity of the full product AB (i.e., $(AB)C = A(BC)$) is the constraint that enforces this joint Lie-Jordan compatibility. This constraint is the algebraic form of the "principle of the now" (Theorem 0.4.4): the requirement that where and when be jointly defined at the same instance.

Definition 0.1.5 — Primitive Constraints

A constraint is an element $\hat{C}_\alpha \in A$ whose vanishing expresses a relational admissibility condition. The framework introduces a collection $\{\hat{C}_\alpha\}_{\alpha \in I}$ of primitive constraints. No first-class/second-class distinction is made at this stage; that distinction requires the compatibility matrix (Definition 0.2.4) and the fracture machinery of Section 1, where the Tier 1 / Tier 2 separation becomes visible.

> **DEFINITION 0.1.5 — Primitive Constraints** [Established]

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The mutual obstruction between constraints is captured by their commutator — the Lie bracket of the constraint algebra. For two primitive constraints $\hat{C}_\alpha, \hat{C}_\beta$, the compatibility bracket is $\Delta_{\{\alpha\beta\}} = [\hat{C}_\alpha, \hat{C}_\beta] \in A$. The full compatibility matrix $\Delta = (\Delta_{\{\alpha\beta\}})$ collects all mutual obstructions.

> **DEFINITION 0.2.4 — Compatibility Bracket** [Established]

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> For primitive constraints $\hat{C}_\alpha, \hat{C}_\beta \in A$, the compatibility bracket is the Lie bracket

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> $\Delta_{\{\alpha\beta\}} = [\hat{C}_\alpha, \hat{C}_\beta] = \hat{C}_\alpha \hat{C}_\beta - \hat{C}_\beta \hat{C}_\alpha \in A$ (0.2.4)

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> The full compatibility matrix $\Delta = (\Delta_{\{\alpha\beta\}})$ collects all mutual Lie-obstructions.

Remark 0.2.5 — The Tier Distinction Is Deferred. The compatibility matrix Δ encodes which constraints commute (first-class, $\Delta_{\{\alpha\beta\}} \approx 0$ modulo constraints) and which do not (second-class, $\Delta_{\{\alpha\beta\}} \neq 0$). This Tier 1 / Tier 2 distinction is not made at Layer L because it requires the fracture machinery of Section 1 to be operationally meaningful: the Dirac reduction (quotienting by the Tier 1 ideal, inverting the Tier 1 compatibility sub-matrix) is performed in Section 1, where the reduced algebra A_{red} and its master constraint $\hat{M}_{\text{red}}$ are defined. At Layer L, Δ is introduced as an algebraic object — the Lie obstruction structure of the constraint algebra; its physical interpretation (the structural origin of the fracture) is a Section 1 result.

Remark 0.2.6 — Lie and Jordan Components of the Constraint Algebra. The constraint algebra inherits both Lie and Jordan structure from A . The compatibility bracket (0.2.4) is the Lie component — it captures ordering obstructions between constraints. The Jordan component — $\hat{C}_\alpha \circ \hat{C}_\beta = \frac{1}{2}(\hat{C}_\alpha \hat{C}_\beta + \hat{C}_\beta \hat{C}_\alpha)$ — captures co-occurrence relations between constraints and plays a role in the sectoral decomposition of Section 1. Both components are present at Layer L; their operational separation (Lie obstructions driving the Dirac reduction, Jordan co-occurrences driving the sectoral decomposition) is a Section 1 development.

§0.3 GNS Construction and the Emergent Hilbert Space (Layer Q — Quadratic)

LAYER Q · QUADRATIC

Source: Front.pdf §0.6, Con.pdf §0.4, Section_0_2 §0.2–0.3. *Epistemic tag:* [Established].

The Quadratic layer introduces the Gel'fand–Naimark–Segal (GNS) construction, which produces a Hilbert-space representation from the kinematic state. The construction is derived, not postulated: H_{kin} is the completion of the quotient of A by the GNS null space, equipped with the state-induced inner product. No primitive Hilbert space is assumed.

This section also introduces the fracture operator $\hat{F}$ — but *only its algebraic definition*. $\hat{F}$ is definable from the compatibility brackets (Definition 0.2.4) without any reference to the

fracture machinery. However, $\hat{F}$'s *physical interpretation* — as the measure of the Hilbert-space fracture, the algebraic signature of the Lie-Jordan mismatch — requires the Open Expansion and fracture constructions of Section 1. We define $\hat{F}$ algebraically here; we interpret it physically in Section 1.

Definition 0.3.1 — State-Induced Inner Product

> **DEFINITION 0.3.1 — State-Induced Inner Product** [Established]

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> Let ω_{kin} be a general kinematic state on A (not required to be admissible). The state-induced sesquilinear form on A is

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$$\triangleright \langle A, B \rangle_\omega := \omega_{\text{kin}}(A \dagger B) \quad \text{for } A, B \in \mathcal{A} \quad (0.3.1)$$
 $\succ$

> This form is linear in the second argument and conjugate-linear in the first. By the positivity of ω_{kin} , it is positive semidefinite: $\langle A, A \rangle_{\omega} = \omega_{\text{kin}}(A^\dagger A) \geq 0$.

Definition 0.3.2 + Lemma 0.3.3 — GNS Null Space

> **DEFINITION 0.3.2 + LEMMA 0.3.3** [Established]

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> The GNS null space is the set of algebra elements of zero state-induced length:

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$$> N_\omega = \{ A \in A : \omega_{\text{kin}}(A^\dagger A) = 0 \} \quad (0.3.2)$$

>

> **Lemma 0.3.3:** N_ω is a left ideal of A .

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> *Proof:* If $A \in N_\omega$ and $B \in A$, then by the Cauchy-Schwarz inequality for positive linear functionals ($|\omega(A^\dagger X)|^2 \leq \omega(A^\dagger A) \cdot \omega(X^\dagger X) = 0$), $\omega_{\text{kin}}(A^\dagger X) = 0$ for all $X \in A$. Set $X = B^\dagger B A$; then $\omega_{\text{kin}}((BA)^\dagger (BA)) = \omega_{\text{kin}}(A^\dagger B^\dagger B A) = 0$, so $BA \in N_\omega$. $\square$

Definition 0.3.4 — Kinematic GNS Hilbert Space

> **DEFINITION 0.3.4 — Kinematic GNS Hilbert Space** [Established]

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> The kinematic GNS Hilbert space H_{kin} is the metric completion of the quotient pre-Hilbert space:

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> H kin = completion of $(A / N \omega, \langle \cdot, \cdot \rangle_\omega)$ (0.3.4)

 $\succ$

> The representation $\pi_{\text{kin}} : A \rightarrow \text{End}(H_{\text{kin}})$ is given by left multiplication: $\pi_{\text{kin}}(A) [B] = [AB]$. The cyclic vector is $\Omega_{\text{kin}} = [1] \in H_{\text{kin}}$, satisfying $\omega_{\text{kin}}(A) = \langle \Omega_{\text{kin}}, \pi_{\text{kin}}(A) \Omega_{\text{kin}} \rangle_{\{H_{\text{kin}}\}}$ (reconstruction).

****Remark 0.3.5 — Lie and Jordan in the GNS Representation.**** The GNS representation π_{kin} carries both the Lie and Jordan structure of A into $\text{End}(H_{\text{kin}})$. The Lie bracket is represented as $[\pi_{\text{kin}}(A), \pi_{\text{kin}}(B)] = \pi_{\text{kin}}([A, B])$; the Jordan product as $\pi_{\text{kin}}(A) \circ$

Definition 0.3.6 — Fracture Operator (Algebraic Definition)

— — —

You're right — and I had it backwards. I was treating the "now" as if it could be defined at §0 on the algebraic primitives (Lie + Jordan = associativity), but that's trying to define the now before either of its constituents exists operationally. The Lie and Jordan structures are the *governing mechanisms* — they determine *how* "when" and "where" work — but they aren't "when" and "where" themselves. "When" is duration, which is built from the causal order (operational $<$) weighted by burden. "Where" is relational/correlational structure, which is built from K_ω and the local sectors. Neither exists at §0. The "now" — the joint definition of where and when — therefore can't be defined at §0 either.

This means the §0 draft has a layering error I need to fix. Here's the corrected structure, with the "now" properly deferred.

The correct layering

Concept	Where it lives	What it is
Lie algebra A_L (ordering structure)	§0	The governing mechanism of "when"
Jordan algebra A_J (co-occurrence structure)	§0	The governing mechanism of "where"
Causality (primitive, the "when"-capacity)	§0	Named via Lie; operationally $<$ built in §1
Locality (primitive, the "where"-capacity)	§0	Named via Jordan; operationally sectors built in §1
Duration (operational "when")	§1 or later	Built from $<$ + burden (the actual "when" mechanism)
Relational/correlational links (operational "where")	§3 (geometry)	Built from K_ω (the actual "where" mechanism)
The "now"* (joint where+when)	§3 or §4 (after both exist)	Defined when space (where) and time (when) are both available
Reconciliation principle (variational)	§2 (after burden)	The dynamic enforcement of now-compatibility
Global desynchronization (mechanism)	§1 (after fracture)	Local nows achieved, globally lagging by finite propagation

The "now" is the joint product of the actual "where" (correlational structure, K_ω) and the actual "when" (duration, burden-weighted causal depth). It can't be defined at §0 because neither constituent exists yet. What CAN be defined at §0 is the *governing mechanism* — Lie and Jordan — which is the algebraic substrate that the actual "where" and "when" will be built from. But the "now" itself is downstream.

What this means for §0

The §0 draft needs two corrections:

****1. Theorem 0.4.4 (Complementarity) must NOT define the "now."***** It should state only the complementarity of the *governing mechanisms* (Lie and Jordan), without claiming that their joint compatibility IS the now. The now is a downstream concept that requires space and time. The theorem should say: Lie and Jordan are independent governing mechanisms; their complementarity is the structural possibility of the now, but the now itself is constructed when space and time are constructed.

****2.** The desynchronization remark must be removed from §0. ****** It references "local nows" and "global lagging," but the now isn't defined yet. The desynchronization mechanism is a §1 (or later) result, stated after the fracture produces sectors and the finite propagation rate is established. At §0, we only name the primitives and their governing mechanisms.

Corrected §0.4 (the only section that changes)

Here's the rewritten §0.4 with the layering fixed. Everything else in the v2.1 draft stands.

§0.4 The Two Structural Primitives: Causality and Locality

****STRUCTURAL PRIMITIVES****

***Source:** New (structural refinement of v1.0 §1.1 and §1.5, grounded in the Lie-Jordan decomposition). ***Epistemic tag:** [Structural — formal characterization; operational definitions constructed in §1 and later].

The GNS construction produces the kinematic Hilbert space H_{kin} . At this point, the framework introduces two structural co-primitives that govern all subsequent emergence: ****causality**** and ****locality****. These are not derived from the algebra; they are the structural principles grounded in the algebra's two independent algebraic structures — the Lie algebra A_L and the Jordan algebra A_J . Their full operational definitions — causality as the strict partial order $<$ on zero-preserving events (the actual "when" mechanism, duration), locality as the sectoral decomposition of H_{kin} and the correlational structure built on it (the actual "where" mechanism, space) — are constructed in later sections, after the machinery that produces events, sectors, and the correlation kernel is available. Here, we name the primitives and give their formal characterizations as governing mechanisms.

The Two-Primitive Picture

In standard physics, causality and locality are typically treated as a single layered structure: spacetime is a manifold equipped with both a causal order (the light-cone structure) and a locality structure (the topology of spatial proximity). This conflation is natural in a background-dependent framework, where spacetime is given primitively. RCF separates them by grounding them in the two independent algebraic structures of A : causality in the Lie algebra (the antisymmetric, ordering structure), locality in the Jordan algebra (the symmetric, co-occurrence structure). Because the Lie and Jordan structures are algebraically independent, causality and locality are genuinely separable co-primitives — neither is inferable from the other.

Definition 0.4.1 — Causality (Primitive)

> ****DEFINITION 0.4.1 — Causality**** [Structural primitive; operational definition in §1]
>

> Causality is the structural principle grounded in the Lie algebra A_L of A . The Lie bracket $[A, B] = AB - BA$ is antisymmetric ($[A, B] = -[B, A]$) and captures *ordering*: the relation $[A, B] \neq 0$ indicates that the order of A and B matters, that there is a directed dependency between them. Causality is the *governing mechanism* of the "when": the capacity of A 's Lie structure to support an irreflexive, asymmetric, transitive dependency relation.

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> Causality determines the structural possibility of *when* admissibility is well-defined: a relational operation B is well-defined only given the prior fixation of its causal antecedent A . The *actual* "when" — duration, the operational ordering of events in sequence — is constructed in Section 4, built from the causal order $<$ (operational, §1) weighted by burden (§1).

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> The operational definition of causality — $<$ as a strict partial order on zero-preserving events — is constructed in Section 1, after events are defined (events require the thin physical sub-algebra $A_{\text{phy}^{\text{thin}}}$, which requires the fracture machinery). Causality as a primitive is the structural possibility of this ordering, available at the foundational layer via the Lie algebra A_L .

Definition 0.4.2 — Locality (Primitive)

> **DEFINITION 0.4.2 — Locality** [Structural primitive; operational definition in §1 and §3]

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> Locality is the structural principle grounded in the Jordan algebra A_J of A . The Jordan product $A \circ B = \frac{1}{2}(AB + BA)$ is symmetric ($A \circ B = B \circ A$) and captures *co-occurrence*: the relation $A \circ B \neq 0$ indicates that A and B are jointly present, independently of any ordering between them. Locality is the *governing mechanism* of the "where": the capacity of A 's Jordan structure to support a commutative co-occurrence relation — the undirected structure of which relational operations are together.

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> Locality determines the structural possibility of *where* admissibility resides: a relational operation B is local to A if they co-occur, independently of any ordering. The *actual* "where" — relational/correlational links, the operational spatial structure — is constructed in Section 3, built from the correlation kernel K_ω on the local sectors.

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> The Jordan structure is state-independent and genuinely independent of the Lie structure (Remark 0.1.3). Neither is inferable from the other — this is the formal content of "you can know where without when, and when without where."

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> The operational definition of locality begins in Section 1 with the sectoral decomposition of H_{kin} into sub-Hilbert spaces (the fracture), and is completed in Section 3 with the construction of the correlation kernel K_ω and the emergent spatial metric. Locality as a primitive is the structural possibility of this decomposition and correlational structure, available at the foundational layer via the Jordan algebra A_J .

Remark 0.4.3 — Locality $\neq$ Local Sectors; Locality $\neq$ Causality's Complement. Locality is not the set of local sectors; it is the *governing mechanism* that generates them, just as causality is the governing mechanism that generates the causal order, not the order itself. The distinction parallels the distinction between causality and time: causality is the primitive

(the Lie structure, the governing mechanism of "when"), time/duration is the emergent quantity built from it (burden-weighted causal depth, constructed in §4). Similarly, locality is the primitive (the Jordan structure, the governing mechanism of "where"), relational/correlational links and space are the emergent structures built from it (the fractured sub-Hilbert spaces of §1 and the correlation geometry of §3). Conflating locality with local sectors — or causality with duration — is a category error that the framework explicitly avoids.

Locality is also not "causality's complement" in the sense of being defined as the absence of causal relation (the antichain reading). Locality is a co-primitive with its own algebraic substrate (the Jordan algebra), genuinely independent of causality (the Lie algebra). Their *complementarity* — the structural possibility of their joint consistency — is a dynamic property (the reconciliation principle), not a definitional one.

Theorem 0.4.4 — Complementarity of the Governing Mechanisms

> ****THEOREM 0.4.4 — Complementarity of the Governing Mechanisms**** [Structural]
>
> Causality (the Lie structure, the governing mechanism of "when") and locality (the Jordan structure, the governing mechanism of "where") are structurally complementary co-primitives. Neither reduces to the other:
>
> (i) The Lie structure $[A,B]$ is antisymmetric and captures ordering. It is the algebraic substrate (governing mechanism) of causality — the "when"-capacity.
>
> (ii) The Jordan structure $A \circ B$ is symmetric and captures co-occurrence. It is the algebraic substrate (governing mechanism) of locality — the "where"-capacity.
>
> (iii) The two are mathematically independent: neither is inferable from the other.
>
> (iv) Their complementarity — the structural possibility of joint "when"/"where" consistency — is enforced by the associativity constraint $AB = \frac{1}{2}[A,B] + A \circ B$, which requires both the Lie and Jordan parts to be jointly specified for any full relation AB .
>
> ****Important scope limitation:**** This theorem establishes the complementarity of the *governing mechanisms* (Lie and Jordan). It does NOT define the "now" — the joint definition of actual "where" (relational/correlational links, constructed in §3) and actual "when" (duration, constructed in §4). The "now" requires both space and time to exist operationally; it is constructed in §4 or later, after both constituents are available. At §0, we establish only that the two governing mechanisms are independent and complementary; the downstream "now" and its dynamic enforcement (the reconciliation principle) are deferred to sections where their constituents exist.

****Remark 0.4.5 — Relation to Standard Physics.**** In standard physics, causality and locality are conflated into a single spacetime structure. This conflation is natural when spacetime is given primitively (as in classical GR or QFT on a fixed background), but it obscures the complementarity between their governing mechanisms and eliminates the structural possibility of a dynamic reconciliation between them. RCF separates the two governing

mechanisms via the Lie-Jordan decomposition. The downstream constructions — duration (actual "when") from the causal order, relational/correlational links (actual "where") from the Jordan-based sectoral structure, and the "now" from their joint definition — are deferred to the sections where their operational machinery is available.

What this changes downstream

The "now" is now properly placed: it lives in §4 (or wherever duration and correlational structure are both first available), not in §0. This means:

****For §1:**** The fracture mechanism is driven by the Lie-Jordan mismatch — the gap between the current Jordan structure's state and the Jordan structure that would be compatible with the current Lie structure. This is still definable at §1, because the Lie and Jordan structures themselves are available at §0. The fracture produces sub-sectors, each with locally-converging Lie-Jordan compatibility. But §1 does NOT define the "now" — it only defines the **local convergence toward Lie-Jordan compatibility**, which is the precursor to the local "now" that gets defined when duration and correlational structure are built.

****For §2:**** The Reconciliation Principle (variational) is the dynamic enforcement of Lie-Jordan compatibility. It can be stated as a variational principle on burden (the cost of the Lie-Jordan mismatch) without referencing the "now." The "now" is the downstream **interpretation** of this enforcement — the production of joint where/when — but the variational principle itself only needs Lie, Jordan, and burden, all of which exist by §1.

****For §3:**** The correlation kernel K_ω and the emergent spatial metric are constructed. This is where the operational "where" (relational/correlational links) first exists.

****For §4:**** Duration (burden-weighted causal depth) is constructed. This is where the operational "when" first exists. With both "where" (§3) and "when" (§4) available, the "now" can be defined as their joint product — and the desynchronization mechanism (local nows achieved, globally lagging by finite propagation) can be formalized. Dark energy, as the residual cost of global desynchronization, is then a §4-or-later result, applied cosmologically in §6.

****For §6 (cosmology):**** The dark energy mechanism references the "now" and its global desynchronization. This is fine, because §6 is downstream of §3 and §4. The dark energy density $\rho_{DE} \sim (H/\Gamma)^2$ with $\Gamma \sim \eta \cdot B$ is the cosmological signature of the desynchronized now — but the "now" itself is defined in §4, not in §6.

The layering test, updated

| Concept | First available at | Built from |

|---|---|---|

| Lie algebra A_L (governing "when"-mechanism) | §0 | Algebraic primitive |

| Jordan algebra A_J (governing "where"-mechanism) | §0 | Algebraic primitive |

| Causality (named primitive) | §0 | Lie algebra |

Locality (named primitive)	§0	Jordan algebra
Lie-Jordan complementarity	§0 (Thm 0.4.4)	Lie + Jordan + associativity
Fracture (sub-Hilbert spaces)	§1	Lie-Jordan mismatch + finite propagation
Operational $<$ (causal order)	§1	Events (from thin A_{phy}) + Lie structure
Burden B_{Δ} (cost of Lie-Jordan mismatch)	§1	$\hat{F}$ (algebraic, §0) + fracture interpretation
Relational/correlational links K_{ω} (operational "where")	§3	Jordan structure + local sectors + GNS state
Duration (operational "when")	§4	$<$ + burden (causal depth weighted by burden density)
**The "now" (joint where+when)	§4	Relational links (§3) + duration (§4)
Reconciliation Principle (variational)	§2	Burden (§1) — formal enforcement of Lie-Jordan compatibility
Global desynchronization mechanism	§1 (fracture) + §4 (now defined)	Local nows + finite propagation
Dark energy (cosmological application)	§6	Desynchronization + cosmology

The "now" is now correctly placed at §4, built from the operational "where" (§3) and the operational "when" (§4). The governing mechanisms (Lie, Jordan) are at §0. The reconciliation principle (§2) enforces Lie-Jordan compatibility without needing to define the now. The desynchronization mechanism straddles §1 (fracture produces local convergence) and §4 (now defined, global lagging formalized). Dark energy (§6) is the cosmological signature.

Status of the §0 draft

The v2.1 draft stands with the §0.4 section replaced by the corrected version above. The acyclicity test still passes — in fact, it's cleaner now, because §0 no longer references the "now" or the desynchronization mechanism at all. Those are properly deferred.

The §0 draft is now:

- §0.0 Purpose (Lie-Jordan as the algebraic substrate; two-primitive picture)
- §0.1 Algebra and Lie-Jordan decomposition (Defs 0.1.1–0.1.7)
- §0.2 Admissibility and constraint ideal (Defs 0.2.1–0.2.4, Remarks 0.2.5–0.2.6)
- §0.3 GNS and $\hat{F}$ algebraic (Defs 0.3.1–0.3.6, Remarks 0.3.5–0.3.7)
- §0.4 Two primitives (Defs 0.4.1–0.4.2, Thm 0.4.4, Remarks 0.4.3, 0.4.5) — **corrected per your fix**
- §0.5 Architectural summary

Ready to proceed to §1 (Open Expansion, Fracture, and the Reconciliation Propagator), or do you want to refine the §0 layering further first?

§0.5 Architectural Summary

What Section 0 Contains

Object	Definition	Layer	Status
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Kinematic algebra A	Def 0.1.1	L	Established
Lie-Jordan decomposition	Def 0.1.2	L	Established
Primitive constraints $\{\hat{C}_\alpha\}$	Def 0.1.5	L	Established
Master constraint $\hat{M}$ (collective)	Def 0.1.6	L	Established
Algebraic state ω	Def 0.2.1	L	Established
Admissibility (zero-constraint compatibility)	Def 0.2.2	L	Established
Constraint ideal I_C	Def 0.2.3	L	Established
Compatibility bracket $\Delta_{\{\alpha\beta\}}$ (Lie obstruction)	Def 0.2.4	L	Established
GNS Hilbert space H_{kin}	Def 0.3.4	Q	Established
Fracture operator $\hat{F}$ (algebraic only)	Def 0.3.6	Q	Established (algebraic); physical interpretation in §1
Causality (Lie primitive)	Def 0.4.1	Structural	Named; operational in §1
Locality (Jordan primitive)	Def 0.4.2	Structural	Named; operational in §1
Complementarity / Principle of the Now	Thm 0.4.4	Structural	Established

What Section 0 Does NOT Contain (Deferred to §1 or Later)

Object	Deferred to	Reason
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Reconciliation Propagator R_t	§1	Requires $\hat{F}$'s physical interpretation (fracture)
Convergence Theorem (Master-Zero derived)	§1	Requires R_t
Thin physical sub-algebra $A_{\text{phy}}^{\text{thin}}$	§1	Requires $\ker(\hat{M})$, which requires the Convergence Theorem
Full physical sub-algebra $A_{\text{phy}}^{\text{full}}$	§1	Requires R_∞
Fracture of H_{kin} into sub-Hilbert spaces	§1	Requires Open Expansion and the Lie-Jordan mismatch
Reduced algebra A_{red}	§1	Requires the fracture (sub-algebras) and the Tier 1/Tier 2 split
Master constraint $\hat{M}_{\text{red}}$ on A_{red}	§1	Requires A_{red} ; defined as the Tier 2 sub-collective of $\hat{M}$
Dirac bracket	§1	Requires A_{red} and the Tier 1 compatibility matrix inversion
Burden B_Δ (as physical quantity)	§1	Requires $\hat{F}$'s physical interpretation
Cross-extension network operator Π_{net}	§1 or later	Requires K_ω (§3) and the sectoral decomposition (§1)
Reconciliation Principle (variational)	§2	Requires burden, which requires the fracture
Zero-preserving events E_{phy}	§1	Requires $A_{\text{phy}}^{\text{thin}}$
Causal order $<$ (operational)	§1	Requires events
Open Expansion Principle	§1	Requires the two named primitives and the Lie-Jordan mismatch
Global desynchronization mechanism	§1	Requires fracture + finite propagation rate

The Emergence Ladder So Far

...

Layer L (Linear):

- A (algebra) $\rightarrow$ Lie-Jordan decomposition (A_L, A_J)
- $\rightarrow \hat{C}_\alpha$ (constraints) $\rightarrow \hat{M}$ (master, collective)
- $\rightarrow \omega$ (state) $\rightarrow$ admissibility $\rightarrow I_C$ (constraint ideal) $\rightarrow \Delta$ (Lie obstruction)

Layer Q (Quadratic):

- GNS $\rightarrow H_{kin}$ (Hilbert space, carrying both Lie and Jordan)
- $\rightarrow \hat{F}$ (fracture operator, algebraic only — measures Lie obstruction)

Structural Primitives:

- Causality (Lie, the "when") + Locality (Jordan, the "where")
- $\rightarrow$ Complementarity = Principle of the Now (associativity $AB = \frac{1}{2}[A, B] + A \circ B$)
- $\rightarrow$ Reconciliation = dynamic production of the now (formalized in §2)
- $\rightarrow$ Local nows achieved, globally desynchronized by finite propagation (mechanism in §1)

...

What This Section Unlocks for Section 1

With Section 0 closed, Section 1 can be merged against a stable foundation. Section 1 will introduce:

1. ****Open Expansion Principle**** — local sectors (built from the Jordan structure, locality) expand to reconcile with the single causal layer (built from the Lie structure, causality) to preserve zero-constraint compatibility.
2. ****Finite propagation rate**** — reconciliation cannot occur instantaneously; this gives the speed limit $c = \gamma \cdot \ell_0$. The Lie structure evolves at rate γ ; the Jordan structure has its own relaxation rate; the mismatch between them drives the fracture.
3. ****The Lie-Jordan mismatch**** — the gap between the current Jordan structure (locality's state) and the Jordan structure that would be perfectly compatible with the current Lie structure (causality's state).
4. ****The fracture**** — the mismatch fractures H_{kin} into sub-Hilbert spaces $\{H_k\}$ and A into sub-algebras $\{A_k\}$. Each sub-sector achieves its own local now; the global now is lagged by finite propagation.
5. **** $\hat{F}$'s physical interpretation**** — the fracture operator measures the Lie-Jordan mismatch. Burden $B_\Delta = \text{Tr}(\rho \hat{F})$ becomes the cost of maintaining the locally-achieved but globally-lagging now.
6. ****The Tier 1 / Tier 2 separation**** — the Lie compatibility matrix Δ reveals which constraints are second-class (Tier 1, quotiented) and which are first-class (Tier 2, surviving). The Jordan co-occurrence structure governs the sectoral decomposition.
7. ****The reduced algebra A_{red} **** — the quotient by the Tier 1 ideal.
8. **** $\hat{M}_{red}$ **** — the Tier 2 sub-collective of $\hat{M}$, living naturally on A_{red} . $\hat{M}$ is not "reduced"; it is **separated** into its Tier 1 and Tier 2 sub-collectives.
9. ****The Dirac bracket**** — the modified commutator on A_{red} , inverting the Tier 1 compatibility sub-matrix.

10. **The Reconciliation Propagator R_t** — the dynamics (Open Expansion $\circ$ Open Extension $\circ$ dephasing) on the fractured structure, driving each local sector toward its own now while propagating reconciliation signals at finite rate.
11. **The Convergence Theorem** — R_t drives any kinematic state to $\ker(\hat{M})$ locally; the global state is a desynchronized composite of locally-converged sectors.
12. **The thin/full split of A_{phy}** — $A_{\text{phy}}^{\text{thin}}$ ($\ker(\hat{M})$ -compatibility) and $A_{\text{phy}}^{\text{full}}$ (R_∞ fixed-point), certified equal by Theorem 0.7.3 (T-2 dependent).

The merger order Section 0 $\rightarrow$ Section 1 is therefore not arbitrary; it is the order in which the fracture machinery becomes available, allowing the objects that were prematurely placed in v1.0 Section 0 to be defined without circularity.

Acyclicity Test

Question: Does Section 0 contain anything that depends on fracture, sub-Hilbert spaces, burden, R_t , A_{red} , $\hat{M}_{\text{red}}$, the Dirac bracket, the Reconciliation Principle, events, the causal order $<$ (operational), Open Expansion, K_ω , or the global desynchronization mechanism?

Answer: No.

- The Lie-Jordan decomposition (Def 0.1.2) is a theorem of the algebra, available at Layer L with no additional structure.
- $\hat{F}$ is defined algebraically (Def 0.3.6) from π_{kin} and Δ , both available at Layer Q. Its physical interpretation is explicitly deferred to §1 (Remark 0.3.7).
- $\text{Tr}(\rho \hat{F})$ is noted as a well-defined non-negative number, but is NOT named "burden." The burden interpretation is deferred to §1.
- Causality and locality are named as primitives with formal characterizations (Defs 0.4.1, 0.4.2) grounded in the Lie and Jordan algebras respectively. Their operational definitions ($<$ on events, sectoral decomposition) are explicitly deferred to §1.
- The complementarity theorem (Thm 0.4.4) names the reconciliation principle as the dynamic consequence of Lie-Jordan complementarity, but does not state it as a variational principle (that is §2, after burden is defined).
- The "now" is characterized algebraically (as the full product AB requiring joint Lie-Jordan compatibility) and physically (as locally achieved, globally lagging), but the *mechanism* of local achievement and global lagging is deferred to §1.
- No forward references to K_ω , Γ_{net} , R_t , A_{red} , $\hat{M}_{\text{red}}$, the Dirac bracket, or the desynchronization machinery.

Verdict: Section 0 is acyclic. The architecture holds at the foundational layer.

End of Section 0 — Rewritten Canonical Form v2.1 (Draft).
